

MATH375 Homework 3

32430286@students.liu.edu.lb [Switch account](#)



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* Indicates required question

Choose the correct answer

The theoretical order of Euler's method is *

- ☐ 3
- ☐ 1
- ☐ 2
- ☐ 4

The theoretical order of modified Euler's method is *

- ☐ 3
- ☐ 2
- ☐ 1
- ☐ 4

*

Consider the first-order IVP:

$$y' = t + y^2, \quad y(0) = 1.$$

Use two-step predictor-corrector Adams-Moulton method with $h = 0.3$, the approximation of $y(0.6)$ using Euler's method to find y_1 is

-0.08222222



- ☐ 0.000000
- ☐ 1.326667
- ☐ 1.9933626
- ☐ 2.2710088

★

Consider the following third-order initial value problem:

$$t^3 y''' - 2t^2 y'' + 3ty' - 3y = 4e^t - 4, \quad y(1) = 1, y'(1) = 0, y''(1) = 1$$

The real numbers a, b, and c such that

$$y(t) = a(3t^3 + 5t) + bt \ln(t) + ct^2$$

is the exact solution of the IVP are:

- ☐ a = 1/4, b = -3/2, c = -1
- ☐ a = 1/4, b = -3/2, c = 1
- ☐ a = 1/4, b = 3/2, c = -1
- ☐ a = -1/4, b = -3/2, c = -1

★

Consider the following first-order initial value problem:

$$y' = 2t(25 - y), \quad y(0) = 1. \quad (E)$$

The approximation of $y(1)$ using Euler's method and the step size $h = 0.5$ is

- ☐ 11
- ☐ 12
- ☐ 13
- ☐ 14

★

Consider the first-order IVP:

$$y'(t) = t\sqrt{y+1}, \quad y(1) = 1.$$

Use modified Euler method with $h = 0.25$, the approximation of $y\left(\frac{3}{2}\right)$ is

- ☐ 7.630686
- ☐ 3.103553
- ☐ 3.672180
- ☐ 6.946935

*

Consider the first-order IVP:

$$y' = t + y^2, \quad y(0) = 1.$$

Use two-step explicit Adams-Bashforth method with $h = 0.1$, the approximation of $y(0.2)$ using Euler's method to find y_1 is

- ☐ 1.2465
- ☐ 1.9651
- ☐ 1.4652
- ☐ 0.1465

*

Consider the following first-order initial value problem:

$$y' = 2t(25 - y), \quad y(0) = 1. \quad (\text{E})$$

If $y(t) = 25 + ce^{-t^2}$ is an exact solution to (E), then the constant c is equal to

- ☐ 2
- ☐ -26
- ☐ -24
- ☐ 26

★

Consider the following third-order initial value problem:

$$t^3 y''' - 2t^2 y'' + 3ty' - 3y = 4e^t - 4, \quad y(1) = 1, \quad y'(1) = 0, \quad y''(1) = 1$$

The approximation of $y'(1.2)$ by using Euler's method with $h = 0.1$ is most nearly:

- ☐ 0.25
- ☐ 0.27
- ☐ 0.26
- ☐ 0.28

★

Consider the first-order IVP:

$$y'(t) = t^2(y^2 - 2y), \quad y(1.5) = 1.$$

The approximation of $y(2)$ using Euler's method and step size $h = 0.25$ is

- ☐ 2.0176
- ☐ 2.0859
- ☐ 1.5625
- ☐ 2.0009

★

Suppose that the truncation error of the modified Euler method has the form

$$c_1 h^2 + c_2 h^3 + \dots$$

Using modified Euler method, the approximations of $y(0.4)$ are

1.8034557 and 1.8167346 with $h = 0.2$ and $h = 0.1$, respectively.

Then the improved approximation of $y(0.4)$ using Richardson's extrapolation is

- ☐ 1.9323418
- ☐ 1.8211609
- ☐ 1.6953679
- ☐ 1.7231417

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Then the improved approximation of $g(0.4)$ using richardson's extrapolation is

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